

**Comparing Mutual Information in Finite Samples:
Wald and Expanded Welch–Satterthwaite
Inference**

Working Draft

Chapters 1–4

Daniel Zhao

Supervisor: Associate Professor Joseph T. Lizier

School of Computer Science
The University of Sydney

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Chapter 1

Introduction

Suppose two independent groups are observed on the same categorical variables X and Y . Mutual information (MI) describes the strength of association between those variables in each group. If the estimated MI differs, the difference alone does not tell us whether the populations differ: both estimates vary when new observations are drawn. The question in this thesis is whether the two *population* MI values can plausibly be equal.

This is not the usual test of whether X and Y are independent within one group. The null hypothesis here permits dependence in both groups and permits their complete joint distributions to differ. It specifies only

$$H_0 : I(P) = I(Q), \quad H_1 : I(P) \neq I(Q), \quad (1.1)$$

where P and Q are the two joint distributions. The samples from P and Q are independent of each other; that sampling assumption does not imply independence of X and Y within either population.

1.1 Why the comparison is difficult

MI is a nonlinear function of all cell probabilities and their row and column margins. With a small sample or rare population cells, an observed table can contain many zeros even though every population probability is positive. The same sampled tables determine both the MI difference and its standard error, so sparsity affects both estimates. A first-order Normal Wald test is simple, but its finite-sample rejection rate need not be close to the nominal level.

Welch–Satterthwaite inference offers a possible adjustment. Instead of comparing the standardised difference with a normal reference, it compares the same statistic

with a Student reference whose effective degrees of freedom describe uncertainty in the estimated variance. For MI, that variance is itself a nonlinear function of the full table. Hutcheson's earlier information-theoretic test assigns each variance component its sample size as degrees of freedom; the familiar ordinary-Welch analogue instead uses $n - 1$. Neither assignment accounts for the way a changed cell also changes the margins and pointwise MI values. The method examined here estimates those degrees of freedom from the sensitivity of the full variance functional.

The resulting reference is an approximation, not a guarantee of better inference. A heavier-tailed reference can reduce excessive false positives, but it can also suppress useful detection when Wald already rejects too rarely. Moreover, one method can fail to produce a valid p-value on samples where the other succeeds. Calibration, detection and valid-result frequency must therefore be considered together.

1.2 Research questions

The thesis asks four related questions. First, how can the sampling uncertainty of the estimated MI variance be turned into an MI-specific Welch–Satterthwaite reference? Second, how does that reference affect false positives, detection and numerical validity relative to Normal Wald? Third, how do table size, margins, baseline dependence, the location of association and unequal sample sizes change the comparison? Fourth, do the observed calibration errors diminish with larger samples, and what additional computation does the correction require?

1.3 Contributions to the field

This thesis contributes to two-sample inference for discrete information measures in four connected ways. The established ingredients are first-order MI inference and Satterthwaite moment matching; the contribution lies in connecting, implementing and testing them for the equal-MI problem.

1. **Mathematical contribution.** The thesis derives the sensitivity of the plug-in MI variance to every cell probability. The derivative retains the induced changes in both margins and in the pointwise MI values, providing the quantities needed to estimate MI-specific component degrees of freedom.
2. **Methodological contribution.** The resulting Expanded Welch procedure turns that variance sensitivity into a usable two-sample test. Its relationship

to the Hutcheson sample-size assignment, the ordinary $n - 1$ assignment and Normal Wald is made explicit, so the effect of the full variance-sensitivity expansion can be distinguished from generic Welch correction; the detailed ablation is retained in the experiment supplement.

3. **Empirical contribution.** A frozen, reproducible experiment evaluates false positives, detection, valid-result frequency and runtime on fixed populations with known MI. It covers 2×2 through 8×8 square tables, two rectangular shapes, unequal samples and stress regimes, while preserving every configuration-level result in a companion atlas instead of replacing the regimes by an overall average.
4. **Practical contribution.** The evidence establishes the limits of a heavier-tailed reference correction. Expanded Welch can improve a liberal Wald test, but it can also worsen an already conservative test and return fewer valid results in sparse samples under the strict implemented component-validity rule. Some of those failures come from the rule rather than the combined formula. Expanded Welch therefore should not replace Wald generally. This negative result identifies when reference-tail adjustment is useful and when bias, sparsity or failure of the first-order approximation requires a different solution.

Taken together, these contributions provide both an explicit MI-specific derivation and a clear decision about its practical value. The thesis does not claim that MI inference, information-theoretic Welch tests or Satterthwaite adjustments are themselves new.

1.4 Scope and organisation

The analysis concerns independent multinomial samples on fixed, aligned categorical alphabets. The principal theory assumes strictly positive cell probabilities and positive first-order MI variances. Exact independence is studied separately as a boundary case, not treated as an ordinary positive-MI regime. The final experiment compares Normal Wald with Expanded Welch at a two-sided nominal level of 0.05. It does not establish performance for paired data, continuous MI estimators or alphabets that grow with sample size.

Chapter 2 supplies the statistical background. Chapters 3 and 4 define the shared statistic and derive the correction. Chapter 5 explains how the populations and samples are generated and how the results are measured. Chapter 6 shows

the complete landscape before the focused comparisons. Chapters 7 and 8 discuss the consequences and the resulting recommendation. The appendices provide the detailed algebra, construction and source-to-result record.

Chapter 2

Background and Literature Review

This chapter develops the concepts needed to compare MI between two categorical populations. It begins with entropy, pointwise MI, and population MI. It then reviews plug-in estimation, first-order inference, the Welch–Satterthwaite approximation, and prior comparisons of information quantities. The final section identifies the gap addressed by the expanded method.

2.1 Information-theoretic foundations

2.1.1 Entropy and information content

For a discrete random variable X with probability mass function $p_X(x)$, the information content of outcome x is $-\log p_X(x)$. Rare outcomes have larger information content because they are less expected. Using natural logarithms gives information in nats. The Shannon entropy is the expected information content,

$$H(X) = - \sum_x p_X(x) \log p_X(x). \quad (2.1)$$

Entropy summarises the uncertainty of one random variable before its value is observed (Shannon, 1948).

For a pair (X, Y) , the joint entropy $H(X, Y)$ is defined from the joint probabilities in the same way. The conditional entropy $H(X | Y) = H(X, Y) - H(Y)$ measures the uncertainty remaining in X after observing Y .

2.1.2 Pointwise and average mutual information

Let $P = (p_{ij})$ be the joint distribution of categorical variables $X \in \{1, \dots, r\}$ and $Y \in \{1, \dots, c\}$. Write

$$p_{i+} = \sum_{j=1}^c p_{ij}, \quad p_{+j} = \sum_{i=1}^r p_{ij} \quad (2.2)$$

for the row and column margins. The pointwise mutual information associated with cell (i, j) is

$$\ell_P(i, j) = \log \left(\frac{p_{ij}}{p_{i+}p_{+j}} \right). \quad (2.3)$$

The denominator is the cell probability under independence with the same margins. A positive value means that the cell occurs more often than independence predicts; a negative value means that it occurs less often.

Mutual information is the probability-weighted average of these pointwise values (Shannon, 1948; Bossomaier et al., 2016):

$$I(P) = \sum_{i=1}^r \sum_{j=1}^c p_{ij} \ell_P(i, j). \quad (2.4)$$

Here the summations are nested: for each row i , the inner sum runs over all columns j . Equivalently, the expression sums once over every cell in the table and may be written with the shorthand $\sum_{i,j}$. Because P is the joint distribution of (X, Y) , the same quantity is conventionally written as $I(X; Y)$. Equivalently,

$$I(X; Y) = H(X) + H(Y) - H(X, Y). \quad (2.5)$$

The quantity is nonnegative and equals zero exactly when $p_{ij} = p_{i+}p_{+j}$ for every cell, including zero-probability cells.

Equation (2.4) is also the Kullback–Leibler divergence between the joint distribution and the product of its margins. MI therefore measures the overall departure from independence, not one particular shape of association.

This scalar summary deliberately discards much of the information in the joint distribution. Two populations can have the same MI while assigning probability to different cells, having different margins, or exhibiting different association patterns. Equality of MI is therefore weaker than equality of distributions. This distinction is central to the present test: the null fixes one scalar functional while leaving the remaining features of each population unrestricted.

The logarithm base determines the reporting unit. This thesis uses natural logarithms, so MI is measured in nats; dividing by $\log 2$ converts to bits. This

conversion rescales MI and its standard error together but leaves the degrees of freedom unchanged, so the test result is identical in either unit.

2.2 Estimating discrete mutual information

Suppose a sample of size n produces counts N_{ij} . The empirical cell probabilities are

$$\hat{p}_{ij} = \frac{N_{ij}}{n}. \quad (2.6)$$

Substitution into Equation (2.4) gives the plug-in estimator

$$\hat{I}(P) = \sum_{i,j:\hat{p}_{ij}>0} \hat{p}_{ij} \log \left(\frac{\hat{p}_{ij}}{\hat{p}_{i+}\hat{p}_{+j}} \right). \quad (2.7)$$

The zero-cell convention follows from $\lim_{x \downarrow 0} x \log x = 0$. Empty cells contribute no direct term, but they still affect the empirical margins and the finite-sample behaviour of the estimator.

Under multinomial sampling, the cell counts are random but constrained to sum to n , so they are dependent. The plug-in estimator remains consistent for a fixed alphabet, but its finite-sample behaviour depends on the complete probability table. In particular, rare cells can change the sampling behaviour even when the alphabet and sample sizes are unchanged.

The plug-in estimator is upward biased: random variation in the cell counts tends to make the empirical table appear more dependent than the population table. For a fixed positive $r \times c$ support, the leading bias is

$$\text{Bias}\{\hat{I}(P)\} = \frac{(r-1)(c-1)}{2n} + O(n^{-2}). \quad (2.8)$$

The factor $(r-1)(c-1)$ increases with table size, while division by n shows that the bias decreases as the sample grows. The bias is therefore largest for larger tables with small samples. The result is closely related to Miller's entropy correction (Miller, 1955), and the same histogram-MI term appears in Moddemeijer (1989). Sparse sampling can produce further error beyond this leading approximation (Paninski, 2003).

Bias matters directly when comparing two populations. If $I(P) = I(Q)$ but $n_P \neq n_Q$, the uncorrected difference has leading bias

$$\frac{(r-1)(c-1)}{2} \left(\frac{1}{n_P} - \frac{1}{n_Q} \right). \quad (2.9)$$

For a 2×2 table with $n_P = 50$ and $n_Q = 100$, Equation (2.9) gives 0.005 nats despite equal population MIs. Chapter 3 corrects this sample-size effect by subtracting $(r - 1)(c - 1)/(2n)$ from each estimate.

2.3 Sampling variance of plug-in MI

To compare two estimated MI values, it is necessary to know how much each estimate would vary across repeated samples. The goal of this section is to derive the large-sample approximation

$$\text{Var}\{\hat{I}(P)\} \approx \frac{V(P)}{n}. \quad (2.10)$$

Here $V(P)$, defined below, measures how much the contribution of one observation varies across the population. The derivation assumes that the population cell probabilities are positive, so the logarithms and their derivatives are well defined. This first-order variance is established for histogram and discrete MI (Moddemeijer, 1989; Brillinger, 2004). The argument proceeds in three steps. First, sampling changes the empirical cell probabilities. Second, a first-order Taylor expansion translates those probability errors into an MI error. Third, that error can be written as a sample average, whose variance is straightforward to calculate.

Step 1: From probability errors to MI error.

Using the entropy identity in Equation (2.5), MI can be written directly in terms of the joint and marginal probabilities:

$$I(P) = \sum_{i,j} p_{ij} \log p_{ij} - \sum_i p_{i+} \log p_{i+} - \sum_j p_{+j} \log p_{+j}. \quad (2.11)$$

Let $f(u) = u \log u$. The derivative needed for its first-order Taylor expansion follows directly from the product rule:

$$f'(u) = \frac{d}{du} \{u \log u\} = \log u + u \frac{1}{u} = \log u + 1. \quad (2.12)$$

The general first-order Taylor formula is

$$f(x) \approx f(a) + f'(a)(x - a), \quad (2.13)$$

where a is the value around which the function is expanded and $x - a$ is the distance from that value. Here the new value is the estimated probability $x = \hat{u}$, the expansion point is the population probability $a = u$, and $f(t) = t \log t$. Therefore,

$x - a = \hat{u} - u$, which is the sampling error in the estimated probability. Substituting these quantities into the general formula gives

$$\hat{u} \log \hat{u} \approx \underbrace{u \log u}_{\text{population value}} + \underbrace{(\log u + 1)(\hat{u} - u)}_{\text{first-order change}}. \quad (2.14)$$

Subtracting the population value from both sides makes the change explicit:

$$\hat{u} \log \hat{u} - u \log u \approx (\log u + 1)(\hat{u} - u). \quad (2.15)$$

Applying this expansion to the three sums in Equation (2.11) gives

$$\begin{aligned} \hat{I}(P) - I(P) &\approx \sum_{i,j} (\log p_{ij} + 1)(\hat{p}_{ij} - p_{ij}) \\ &\quad - \sum_i (\log p_{i+} + 1)(\hat{p}_{i+} - p_{i+}) \\ &\quad - \sum_j (\log p_{+j} + 1)(\hat{p}_{+j} - p_{+j}). \end{aligned} \quad (2.16)$$

The margin errors are sums of the corresponding cell errors:

$$\hat{p}_{i+} - p_{i+} = \sum_j (\hat{p}_{ij} - p_{ij}), \quad \hat{p}_{+j} - p_{+j} = \sum_i (\hat{p}_{ij} - p_{ij}). \quad (2.17)$$

For example, the row-margin sum can now be written as

$$\begin{aligned} \sum_i (\log p_{i+} + 1)(\hat{p}_{i+} - p_{i+}) &= \sum_i \left[(\log p_{i+} + 1) \left\{ \sum_j (\hat{p}_{ij} - p_{ij}) \right\} \right] \\ &= \sum_{i,j} (\log p_{i+} + 1)(\hat{p}_{ij} - p_{ij}). \end{aligned} \quad (2.18)$$

The inner sum over j is the probability error for a fixed row i . It is multiplied by that row's coefficient $\log p_{i+} + 1$; the outer sum then adds these products over all rows. The column-margin sum can be rearranged in the same way. Consequently, the same cell error $\hat{p}_{ij} - p_{ij}$ appears in the joint term, its row-margin term and its column-margin term. Its combined coefficient is

$$\begin{aligned} &(\log p_{ij} + 1) - (\log p_{i+} + 1) - (\log p_{+j} + 1) \\ &= \log \left(\frac{p_{ij}}{p_{i+} p_{+j}} \right) - 1 \\ &= \ell_P(i, j) - 1. \end{aligned} \quad (2.19)$$

Therefore,

$$\hat{I}(P) - I(P) \approx \sum_{i,j} \{\ell_P(i,j) - 1\}(\hat{p}_{ij} - p_{ij}). \quad (2.20)$$

The empirical and population cell probabilities both sum to one, so

$$\sum_{i,j} (\hat{p}_{ij} - p_{ij}) = \sum_{i,j} \hat{p}_{ij} - \sum_{i,j} p_{ij} = 1 - 1 = 0. \quad (2.21)$$

The constant -1 in Equation (2.19) therefore contributes zero. Explicitly,

$$\begin{aligned} \hat{I}(P) - I(P) &\approx \sum_{i,j} \{\ell_P(i,j) - 1\}(\hat{p}_{ij} - p_{ij}) \\ &= \sum_{i,j} \ell_P(i,j)(\hat{p}_{ij} - p_{ij}) - \sum_{i,j} (\hat{p}_{ij} - p_{ij}) \\ &= \sum_{i,j} \ell_P(i,j)(\hat{p}_{ij} - p_{ij}). \end{aligned} \quad (2.22)$$

The complicated changes in the joint table and its two margins have therefore reduced to a weighted sum of cell-probability errors. The weight for a cell is its population pointwise MI.

Step 2: From cell errors to observation-level errors.

Let $Z_a = (X_a, Y_a)$ be observation a . An empirical cell probability is the proportion of observations in that cell, so

$$\hat{p}_{ij} - p_{ij} = \frac{1}{n} \sum_{a=1}^n \{\mathbf{1}(Z_a = (i,j)) - p_{ij}\}. \quad (2.23)$$

Substituting this expression into Equation (2.22) first gives

$$\hat{I}(P) - I(P) \approx \sum_{i,j} \ell_P(i,j) \left[\frac{1}{n} \sum_{a=1}^n \{\mathbf{1}(Z_a = (i,j)) - p_{ij}\} \right]. \quad (2.24)$$

The finite sums can be exchanged, so the contribution from observation a is gathered inside one pair of brackets:

$$\begin{aligned} \hat{I}(P) - I(P) &\approx \frac{1}{n} \sum_{a=1}^n \left[\sum_{i,j} \ell_P(i,j) \mathbf{1}(Z_a = (i,j)) - \sum_{i,j} p_{ij} \ell_P(i,j) \right] \\ &= \frac{1}{n} \sum_{a=1}^n \{\ell_P(Z_a) - I(P)\}. \end{aligned} \quad (2.25)$$

The first sum inside the brackets selects the single cell containing Z_a , giving $\ell_P(Z_a)$. The second is the probability-weighted average of all pointwise-MI values, which is $I(P)$. This equation is the key step. To first order, the error in the plug-in

MI estimate behaves like the average of n observation-level contributions. The contribution of an observation in cell (i, j) is simply that cell's pointwise MI minus the population average, $\ell_P(i, j) - I(P)$. Its mean is zero because

$$\begin{aligned} E_P\{\ell_P(X, Y) - I(P)\} &= \sum_{i,j} p_{ij} \{\ell_P(i, j) - I(P)\} \\ &= \underbrace{\sum_{i,j} p_{ij} \ell_P(i, j)}_{I(P)} - I(P) \underbrace{\sum_{i,j} p_{ij}}_1 = 0. \end{aligned} \quad (2.26)$$

In formal asymptotic terminology, the centred function $\ell_P(i, j) - I(P)$ is the influence function of MI; Equation (2.25) is its first-order linear representation (Hampel, 1974; van der Vaart, 1998; Kandasamy et al., 2015).

Step 3: Calculate the sampling variance.

Define the population variance of these centred contributions by

$$V(P) = \text{Var}_P\{\ell_P(X, Y) - I(P)\} = \sum_{i,j} p_{ij} \{\ell_P(i, j) - I(P)\}^2. \quad (2.27)$$

Under independent sampling, the n terms in Equation (2.25) are independent and each has variance $V(P)$. The covariance between any two different observations is therefore zero. Hence

$$\begin{aligned} \text{Var}\{\hat{I}(P)\} &\approx \text{Var}\left\{\frac{1}{n} \sum_{a=1}^n \{\ell_P(Z_a) - I(P)\}\right\} \\ &= \frac{1}{n^2} \sum_{a=1}^n \text{Var}\{\ell_P(Z_a) - I(P)\} \\ &= \frac{1}{n^2} \sum_{a=1}^n V(P) = \frac{nV(P)}{n^2} = \frac{V(P)}{n}. \end{aligned} \quad (2.28)$$

This proves the approximation stated in Equation (2.10). It is the familiar result that the variance of an average of n independent contributions is the variance of one contribution divided by n .

When $V(P) > 0$, the central limit theorem also gives

$$\frac{\hat{I}(P) - I(P)}{\sqrt{V(P)/n}} \xrightarrow{d} N(0, 1). \quad (2.29)$$

This does not claim that every finite-sample MI estimate is normally distributed. It is a local, large-sample approximation for a fixed table with positive population cell probabilities. The $1/n$ leading bias from Section 2.2 decreases faster than the $1/\sqrt{n}$ sampling error, so it does not change this limiting distribution.

The independence boundary.

At exact independence, $p_{ij} = p_{i+}p_{+j}$, so $\ell_P(i, j) = \log 1 = 0$ in every cell. Hence $I(P) = 0$, $V(P) = 0$, and the first-order average in Equation (2.25) vanishes.

A finite sample can still have positive MI. Its leading variation comes from the squared cell-probability errors in a second-order Taylor expansion, which produces the usual chi-squared limit. The normal limit for comparing positive MI values and the chi-squared limit for testing independence therefore apply in different settings.

2.4 Related inferential methods

Normal Wald inference.

A Wald test estimates a parameter difference, divides it by an estimated standard error, and compares the result with its asymptotic normal reference. For two independent MI estimates, the two first-order variance contributions add:

$$\text{Var}\{\hat{I}(P) - \hat{I}(Q)\} \approx \frac{V(P)}{n_P} + \frac{V(Q)}{n_Q}. \quad (2.30)$$

Each term is the sampling variance of one MI estimate. Independence of the two samples makes their covariance zero. Replacing the population variances by estimates gives the standardisation

$$\frac{\hat{I}(P) - \hat{I}(Q)}{\sqrt{\hat{V}(P)/n_P + \hat{V}(Q)/n_Q}} \xrightarrow{d} N(0, 1) \quad \text{under } I(P) = I(Q). \quad (2.31)$$

Here $\hat{V}(P)$ and $\hat{V}(Q)$ estimate the pointwise-MI variances $V(P)$ and $V(Q)$. This limit assumes the regular positive-support conditions of Section 2.3 and increasing sample sizes. Chapter 3 applies the leading bias correction to the numerator and specifies the full Normal Wald test.

Wald inference is asymptotically valid under regularity conditions, but its finite-sample calibration depends on both the quality of the linear expansion and the stability of the estimated standard error. A normal reference does not separately represent the uncertainty of the two estimated variance contributions.

The Welch–Satterthwaite principle.

Welch’s test compares two means when their population variances may differ (Welch, 1947). For two independent samples with means $\bar{X}_P, \bar{X}_Q$ and variances s_P^2, s_Q^2 , the statistic for testing equal population means and its effective degrees of freedom are

$$T_{\text{Welch}} = \frac{\bar{X}_P - \bar{X}_Q}{\sqrt{s_P^2/n_P + s_Q^2/n_Q}}, \quad (2.32)$$

$$\nu = \frac{\left(s_P^2/n_P + s_Q^2/n_Q\right)^2}{\frac{(s_P^2/n_P)^2}{n_P - 1} + \frac{(s_Q^2/n_Q)^2}{n_Q - 1}}. \quad (2.33)$$

The denominator of the statistic contains two independently estimated variance contributions. The degrees-of-freedom formula weights their uncertainty: under normal sampling, each sample variance has component degrees of freedom $n_P - 1$

or $n_Q - 1$. Satterthwaite obtains the combined value by representing the sum as a scaled chi-squared variable with matching mean and variance (Satterthwaite, 1946).

Welch evaluates the statistic against a Student distribution with ν degrees of freedom. The connection to the normal reference can be seen from the representation

$$t_\nu = \frac{Z}{\sqrt{\chi_\nu^2/\nu}}, \quad Z \sim N(0, 1), \quad (2.34)$$

where the normal and chi-squared variables are independent. The denominator is random, which produces heavier tails than the normal reference. As ν increases, χ_ν^2/ν approaches one and the Student reference approaches $N(0, 1)$.

For MI, the estimated variance contributions are $\hat{V}(P)/n_P$ and $\hat{V}(Q)/n_Q$. Their component degrees of freedom need their own calculation because $\hat{V}(P)$ and $\hat{V}(Q)$ are nonlinear functions of the probability tables. Chapter 4 derives these MI-specific inputs to the same moment-matching structure.

For a single scalar restriction, normal and chi-squared Wald notation are equivalent. If T is asymptotically standard normal, then T^2 is asymptotically χ_1^2 . The signed form is retained here because the parameter is the scalar difference $I(P) - I(Q)$, its sign is meaningful, and the Welch refinement replaces the normal reference for T by a Student reference. The $(r - 1)(c - 1)$ -degree chi-squared law used for one-table independence is a different second-order result.

The same principle extends beyond ordinary sample variances. Pan and Wall (2002) account for variability in sandwich variance estimates using approximate Student and F references. Bell and McCaffrey (2002) use Satterthwaite moment matching for regression variance estimates under multi-stage sampling after showing that traditional linearisation estimators can have substantial negative bias. Kauermann and Carroll (2001) examine the variability and efficiency of sandwich covariance estimation and propose an adjustment for the resulting confidence-interval undercoverage. Related adjustments and practical guidance are given by Fay and Graubard (2001) and Imbens and Kolesár (2016). These are close methodological precedents: estimating the variability of a variance estimator and using it in a finite-sample reference are established strategies. The present task is to derive the required inputs for the discrete MI functional and assess whether the resulting reference is useful.

The direction of the finite-sample problem is nevertheless different here. The cited regression corrections respond to underestimated uncertainty or anti-conservative coverage, so a heavier-tailed reference can move inference in the required direction. Near independence, the plug-in MI variance can instead be biased upward. A

heavier-tailed Student reference may then reinforce, rather than offset, the resulting conservatism. This possibility is derived more precisely in Chapter 4 and evaluated in the experiments.

Prior MI inference and related comparisons.

MI uncertainty also has a broader literature than its leading normal approximation. Roulston (1999) derives entropy and MI error estimates using error propagation and examines them numerically. Hutter (2002) and Hutter and Zaffalon (2005) derive moments of the posterior distribution of MI under a Dirichlet model, including higher-order variance and shape information. Their posterior uncertainty has a different conditioning from the repeated-sampling distribution studied here. Their results establish that higher-order MI uncertainty is not a new topic, but do not directly validate the two-sample Student reference in this thesis.

Hutcheson’s test is a direct information-theoretic Welch predecessor. It compares two Shannon diversity estimates with an estimated variance and a Student reference, using

$$\nu_H = \frac{(s_{H,P}^2 + s_{H,Q}^2)^2}{(s_{H,P}^2)^2/n_P + (s_{H,Q}^2)^2/n_Q} \quad (2.35)$$

as its effective degrees of freedom (Hutcheson, 1970). Here each of $s_{H,P}^2$ and $s_{H,Q}^2$ is the estimated sampling variance of the corresponding entropy estimator $\widehat{H}$, and each variance component is assigned its sample size, not $n_i - 1$. Applying the same assignment to the two MI variance contributions gives a clear historical baseline. It does not, however, derive how the plug-in MI variance itself fluctuates: MI depends on a joint distribution and both margins, so moving mass into one cell changes pointwise values elsewhere in its row and column. The expanded calculation retains these changes.

The question of comparing MI values also has predecessors. In a segregation application, Mora and Ruiz-Castillo (2009) develop inference for an MI index and discuss pairwise comparisons across groups and periods; the later peer-reviewed account places that MI index within the wider segregation literature (Mora and Ruiz-Castillo, 2011). They do not derive the variance-sensitivity degrees of freedom used here, but the general aim of comparing MI values is not new. Guha and Chothia (2014) use MI for a different two-sample question: whether two continuous distributions are identical. Their MI is between a pooled observation and its sample label; their null is equality of distributions, not equality of two within-population MI values. For discrete data, Drake and Guha (2014) develop an MI-based k -sample test for the same distributional-equality question.

The exact-independence boundary has likewise been studied separately. Mari-

nescu and Balcau (2025) use a second-order delta-method argument for MI under independence. More generally, their expansion combines a normal term with a linear combination of chi-squared terms; the normal term vanishes at exact independence. That setting explains why a first-order positive-MI comparison must not be presented as a replacement for a one-table independence test and suggests why second-order effects can remain relevant near that boundary.

Table 2.1 distinguishes the relevant established ideas by the problem each addresses. This is not a claim that no other MI comparison has been proposed. It states which pieces are used here and why the final comparison still needs empirical validation.

Table 2.1: Selected prior work, grouped by the question it addresses.

Topic	Established result and relation to this thesis
MI estimation	Miller (1955) and Paninski (2003) establish bias and sparse-sample limitations of plug-in information estimates.
MI variance	Moddemeijer (1989) and Brillinger (2004) give predecessors for the first-order MI variance, while Moddemeijer (1999) treats dependent observation pairs. This thesis instead studies the sampling uncertainty of the plug-in variance estimate itself in the independent two-sample problem.
Welch principle	Satterthwaite (1946) and Welch (1947) provide moment-matching ideas; Hutcheson (1970) applies the same structure to Shannon diversity using component degrees of freedom n_i .
Other comparisons	Mora and Ruiz-Castillo (2009, 2011) study an MI-based segregation index. They do not derive the variance-sensitivity degrees of freedom used here.
Different MI nulls	Guha and Chothia (2014) and Drake and Guha (2014) test distributional equality; Marinescu and Balcau (2025) study one-table independence. These are not the two-population equal-MI null.
Weak-null tests	Chung and Romano (2013) explain why a raw pooled-label permutation need not be valid when the compared parameters agree but $P \neq Q$; Janssen (1997) gives an earlier studentized permutation result for heterogeneous two-sample nulls.

Resampling and the weak null.

Bootstrap methods can approximate a statistic’s repeated-sampling behaviour by drawing from fitted empirical populations, at greater computational cost than a direct analytic table scan (Efron, 1979). Sparse empirical support can still be a problem. A label permutation instead pools the observations and reallocates their group labels; it targets a different symmetry assumption.

Permutation requires a separate distinction. Pooling two samples and permuting their group labels is exact under the strong null $P = Q$, because the observations are exchangeable. Under the weak null $I(P) = I(Q)$, the two complete distributions may differ, so raw label permutation generally targets the wrong null. Studentized permutation can recover asymptotic validity for some weak-null parameter comparisons (Janssen, 1997; Chung and Romano, 2013), but it is not the standard one-table MI shuffle used to test independence.

The final comparison uses Normal Wald and the proposed Expanded Welch method, which share the same effect estimate and standard error. Resampling is relevant future work, but is not an exact finite-sample solution for the unrestricted weak null without additional assumptions.

2.5 Literature synthesis and methodological gap

The literature establishes all of the broad ingredients used in this thesis: plug-in MI estimation and bias correction, first-order MI variance, Wald comparison of smooth functionals, and Welch–Satterthwaite moment matching. It also contains prior comparisons of entropy and MI-related indices.

This thesis makes three contributions to two-sample inference for mutual information. First, it derives an explicit observation-level sensitivity for the estimated MI variance. Second, it uses this sensitivity to construct a plug-in Welch–Satterthwaite degrees-of-freedom correction. Third, it provides a controlled empirical evaluation across distinct equal-MI multinomial populations. Together, these contributions clarify when Welch–Satterthwaite correction is useful for MI comparison and, importantly, why it can become overly conservative.

The remaining task is specific to MI. Hutcheson’s direct information-theoretic predecessor assigns each component its sample size; an ordinary-Welch analogue uses $n - 1$. Both are fixed assignments. By contrast, established sandwich adjustments derive effective degrees of freedom from the estimated variability of the variance estimator itself. The plug-in MI variance requires its own version of that calculation because it is a nonlinear function of the same empirical joint table used to calculate MI. A change in one observation modifies the cell probabilities, margins, pointwise-MI values, and their weighted variance.

The expanded method addresses this gap by differentiating the full MI variance functional. The resulting observation-level sensitivity determines the sampling variance of the variance estimate and, through Satterthwaite moment matching, an MI-specific component degree of freedom. Chapters 3 and 4 develop this

construction, and Chapter 6 evaluates its practical behaviour.

Chapter 3

Comparing Mutual Information with Normal Wald

This chapter defines the estimand and statistic shared by the two methods in the final experiment. Normal Wald evaluates it against a standard normal reference; Chapter 4 develops a Student reference with estimated degrees of freedom.

3.1 Problem setup and bias-corrected estimator

Let

$$N^P = (N_{ij}^P), \quad N^Q = (N_{ij}^Q) \quad (3.1)$$

be independent $r \times c$ count tables with totals n_P and n_Q . The row and column categories have the same definitions in both populations. The empirical probabilities are

$$\hat{p}_{ij} = \frac{N_{ij}^P}{n_P}, \quad \hat{q}_{ij} = \frac{N_{ij}^Q}{n_Q}. \quad (3.2)$$

The population parameter is the signed difference

$$\Delta = I(P) - I(Q). \quad (3.3)$$

The principal test is two-sided:

$$H_0 : \Delta = 0 \quad \text{against} \quad H_1 : \Delta \neq 0. \quad (3.4)$$

The first-order construction uses the following assumptions.

Assumption 1 (Sampling). *Observations are independent and identically distributed within each population, and the two population samples are independent.*

Assumption 2 (Aligned fixed alphabets). *The row and column categories are fixed in advance and aligned between the two tables.*

Assumption 3 (Positive support). *Every population cell has positive probability. Empirical zero cells are allowed, but widespread observed support loss is recorded as a diagnostic.*

Assumption 4 (Regular MI). *The first-order influence variances $V(P)$ and $V(Q)$ are positive. This excludes exact independence and other degenerate cases from the regular first-order approximation.*

All logarithms are natural. MI differences and standard errors are therefore reported in nats. Changing the logarithm base scales both by the same positive constant and leaves the standardised statistic unchanged.

Bias-corrected estimator.

For the first table, define empirical pointwise MI by

$$\hat{\ell}_P(i, j) = \log \left(\frac{\hat{p}_{ij}}{\hat{p}_{i+}\hat{p}_{+j}} \right) \quad \text{when } \hat{p}_{ij} > 0. \quad (3.5)$$

The plug-in MI estimate is

$$\hat{I}(P) = \sum_{i,j:\hat{p}_{ij}>0} \hat{p}_{ij} \hat{\ell}_P(i, j). \quad (3.6)$$

The same calculation gives $\hat{I}(Q)$.

Under fixed positive support, subtracting the leading bias from each plug-in estimate gives the estimated effect directly:

$$\hat{\Delta}_{\text{BC}} = \left\{ \hat{I}(P) - \frac{(r-1)(c-1)}{2n_P} \right\} - \left\{ \hat{I}(Q) - \frac{(r-1)(c-1)}{2n_Q} \right\}. \quad (3.7)$$

The two terms in braces are the bias-corrected MI estimates for the separate populations.

The correction uses the declared alphabet dimensions, not the observed nonempty dimensions. A corrected estimate can be negative in a sparse sample and is not clipped at zero, since clipping would define a different nonlinear estimator. When $n_P = n_Q$, the two corrections cancel in the difference; otherwise, their difference removes the leading unequal-sample bias described in Section 2.2.

3.2 Shared standard error and test statistic

Section 2.3 showed that the first-order sampling variance of $\hat{I}(P)$ is $V(P)/n_P$, where $V(P)$ is the population variance of pointwise MI. Subtracting the fixed leading bias term in Equation (3.7) does not change this variance.

The observed table estimates $V(P)$ by

$$\hat{V}(P) = \sum_{i,j:\hat{p}_{ij}>0} \hat{p}_{ij} \{\hat{\ell}_P(i,j) - \hat{I}(P)\}^2. \quad (3.8)$$

Define $\hat{V}(Q)$ analogously.

Because the two samples are independent, their estimated variance contributions add:

$$\widehat{\text{SE}}^2 = \frac{\hat{V}(P)}{n_P} + \frac{\hat{V}(Q)}{n_Q}. \quad (3.9)$$

The common standardised statistic is

$$T = \frac{\hat{\Delta}_{\text{BC}}}{\widehat{\text{SE}}}. \quad (3.10)$$

The null hypothesis is not imposed when calculating $\hat{\Delta}_{\text{BC}}$, $\widehat{\text{SE}}$ or T : they use the observed tables. Equal population MI enters when interpreting the reference distribution, which is centred at zero under H_0 . Equations (3.7)–(3.10) are shared by Normal Wald and Expanded Welch. Their difference lies in the reference distribution, not the effect estimate or plug-in standard error.

3.3 Normal Wald reference and motivation for expansion

Under the regular first-order asymptotics,

$$T \xrightarrow{d} N(0,1) \quad \text{under } H_0. \quad (3.11)$$

Here both n_P and n_Q increase while the aligned alphabet and the two positive-support population tables remain fixed. The two-sided Normal Wald p-value is

$$p_{\text{Wald}} = 2\{1 - \Phi(|T|)\}, \quad (3.12)$$

where Φ is the standard normal distribution function. The thesis evaluates this test through p-values and rejection rates. It does not claim confidence-interval coverage,

which would require a separate coverage experiment.

The normal reference is a large-sample approximation. Wald estimates the standard error from the tables but applies no degrees-of-freedom adjustment.

Why consider an expanded reference?

The shared statistic supplies the estimated effect and its first-order standard error. The direct Hucheson-style information-theoretic analogue assigns component degrees of freedom n_P and n_Q ; a familiar ordinary-Welch analogue uses $n_P - 1$ and $n_Q - 1$. For positive variance contributions, their combined degree of freedom is at least $\min(n_P, n_Q) - 1$, so at sample sizes of 50 or more their Student reference is already close to normal. The supplementary ablation retains these baselines alongside Expanded Welch. However, $\hat{V}(P)$ is a nonlinear functional of a table whose margins and pointwise MI values also change with the data. The expanded method instead estimates component degrees of freedom from that full functional.

Satterthwaite moment matching requires the first two moments of each positive variance estimate. To first order,

$$\mathbb{E}\{\hat{V}(P)\} \approx V(P). \quad (3.13)$$

Chapter 4 derives $\text{Var}\{\hat{V}(P)\}$ by calculating how the full variance functional changes when one observation changes the joint table.

Chapter 4

Expanded Welch–Satterthwaite Method

The expanded method retains the bias-corrected difference, standard error, and test statistic from Chapter 3. It evaluates that statistic against an estimated Student reference rather than the standard normal. This chapter derives the required degrees of freedom from the sampling variability of the full MI variance estimator.

4.1 From Welch–Satterthwaite to the required MI quantity

For the two MI estimates, the Welch–Satterthwaite equation combines their estimated variance contributions as

$$\nu = \frac{\left\{ \hat{V}(P)/n_P + \hat{V}(Q)/n_Q \right\}^2}{\left\{ \hat{V}(P)/n_P \right\}^2 / \nu_P + \left\{ \hat{V}(Q)/n_Q \right\}^2 / \nu_Q}. \quad (4.1)$$

The numerator contains the two terms in the squared standard error. The remaining quantities are the component degrees of freedom ν_P and ν_Q , which should measure how reliably $\hat{V}(P)$ and $\hat{V}(Q)$ estimate their population counterparts.

The derivation proceeds in four stages. First, Satterthwaite moment matching expresses ν_P in terms of the mean and variance of $\hat{V}(P)$. Second, a one-cell perturbation gives the observation-level sensitivity of $V(P)$. Third, the variability of these sensitivities gives the sampling variance of $\hat{V}(P)$. Finally, the two estimated component degrees of freedom are combined through Equation (4.1).

Moment matching for one component.

Satterthwaite represents a positive variance estimate by a scaled chi-squared variable. For the first population, the working approximation represents $\hat{V}(P)$ by

$$\mathbb{E}\{\hat{V}(P)\} \frac{\chi_{\nu_P}^2}{\nu_P}. \quad (4.2)$$

The approximation matches moments; it does not assert that the finite-sample distribution of $\hat{V}(P)$ is exactly chi-squared.

For a chi-squared variable with ν_P degrees of freedom,

$$\mathbb{E}(\chi_{\nu_P}^2) = \nu_P, \quad \text{Var}(\chi_{\nu_P}^2) = 2\nu_P. \quad (4.3)$$

The scaling in Equation (4.2) therefore matches the mean automatically:

$$\mathbb{E} \left[\mathbb{E}\{\hat{V}(P)\} \frac{\chi_{\nu_P}^2}{\nu_P} \right] = \mathbb{E}\{\hat{V}(P)\}. \quad (4.4)$$

Its variance is

$$\text{Var} \left[\mathbb{E}\{\hat{V}(P)\} \frac{\chi_{\nu_P}^2}{\nu_P} \right] = \frac{[\mathbb{E}\{\hat{V}(P)\}]^2}{\nu_P^2} \text{Var}(\chi_{\nu_P}^2) \quad (4.5)$$

$$= \frac{2[\mathbb{E}\{\hat{V}(P)\}]^2}{\nu_P}. \quad (4.6)$$

Matching this quantity to $\text{Var}\{\hat{V}(P)\}$ gives

$$\boxed{\nu_P = \frac{2[\mathbb{E}\{\hat{V}(P)\}]^2}{\text{Var}\{\hat{V}(P)\}}}. \quad (4.7)$$

To first order, $\mathbb{E}\{\hat{V}(P)\} \approx V(P)$. The required calculation is therefore the sampling variance of $\hat{V}(P)$.

4.2 Deriving the sensitivity of the MI variance

The sampling behaviour of a smooth plug-in functional can be obtained by measuring its first-order response to one observation. Fix a cell (x, y) and define the perturbed distribution

$$P_\varepsilon = (1 - \varepsilon)P + \varepsilon\delta_{(x,y)}, \quad (4.8)$$

where $\delta_{(x,y)}$ places probability one on cell (x, y) . At $\varepsilon = 0$, the distribution is P . Increasing ε moves a small amount of probability toward the selected cell while preserving total probability one.

For an arbitrary cell (i, j) ,

$$p_{ij}(\varepsilon) = (1 - \varepsilon)p_{ij} + \varepsilon \mathbf{1}\{i = x, j = y\}. \quad (4.9)$$

The joint and marginal probability derivatives at the original distribution are

$$\left. \frac{dp_{ij}(\varepsilon)}{d\varepsilon} \right|_0 = \mathbf{1}\{i = x, j = y\} - p_{ij}, \quad (4.10)$$

$$\left. \frac{dp_{i+}(\varepsilon)}{d\varepsilon} \right|_0 = \mathbf{1}\{i = x\} - p_{i+}, \quad (4.11)$$

$$\left. \frac{dp_{+j}(\varepsilon)}{d\varepsilon} \right|_0 = \mathbf{1}\{j = y\} - p_{+j}. \quad (4.12)$$

The derivatives of the joint probabilities sum to zero because

$$\sum_{i,j} \{\mathbf{1}(i = x, j = y) - p_{ij}\} = 1 - 1 = 0. \quad (4.13)$$

Effect on pointwise MI and MI.

Along the perturbation path, pointwise MI is

$$\ell_{P_\varepsilon}(i, j) = \log p_{ij}(\varepsilon) - \log p_{i+}(\varepsilon) - \log p_{+j}(\varepsilon). \quad (4.14)$$

Differentiating the three logarithms and substituting the probability derivatives gives

$$\left. \frac{d}{d\varepsilon} \ell_{P_\varepsilon}(i, j) \right|_0 = \frac{\mathbf{1}\{i = x, j = y\}}{p_{ij}} - \frac{\mathbf{1}\{i = x\}}{p_{i+}} - \frac{\mathbf{1}\{j = y\}}{p_{+j}} + 1. \quad (4.15)$$

MI along the path is

$$I(P_\varepsilon) = \sum_{i,j} p_{ij}(\varepsilon) \ell_{P_\varepsilon}(i, j). \quad (4.16)$$

The product rule separates its derivative into the change in probability weights and the change in pointwise-MI values:

$$\begin{aligned} \left. \frac{dI(P_\varepsilon)}{d\varepsilon} \right|_0 &= \sum_{i,j} \{\mathbf{1}(i = x, j = y) - p_{ij}\} \ell_P(i, j) \\ &\quad + \sum_{i,j} p_{ij} \left. \frac{d\ell_{P_\varepsilon}(i, j)}{d\varepsilon} \right|_0. \end{aligned} \quad (4.17)$$

The first sum is $\ell_P(x, y) - I(P)$. Substitution of Equation (4.15) into the second

sum gives $1 - 1 - 1 + 1 = 0$. Hence

$$\left. \frac{dI(P_\varepsilon)}{d\varepsilon} \right|_0 = \ell_P(x, y) - I(P). \quad (4.18)$$

This result recovers the centred pointwise-MI influence function used in Chapter 3.

Effect on the MI variance.

The target is the first-order change in

$$V(P) = \text{Var}_P\{\ell_P(X, Y)\}. \quad (4.19)$$

Write this variance in second-moment form:

$$V(P) = \mathbb{E}_P\{\ell_P(X, Y)^2\} - I(P)^2. \quad (4.20)$$

Define the observation-level variance sensitivity by

$$g_P(x, y) = \left. \frac{dV(P_\varepsilon)}{d\varepsilon} \right|_0. \quad (4.21)$$

Differentiating Equation (4.20) gives

$$g_P(x, y) = \left. \frac{d}{d\varepsilon} \mathbb{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X, Y)^2\} \right|_0 - 2I(P) \left. \frac{dI(P_\varepsilon)}{d\varepsilon} \right|_0. \quad (4.22)$$

It remains to differentiate

$$\mathbb{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X, Y)^2\} = \sum_{i,j} p_{ij}(\varepsilon) \ell_{P_\varepsilon}(i, j)^2. \quad (4.23)$$

The product rule gives

$$\begin{aligned} \left. \frac{d}{d\varepsilon} \mathbb{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X, Y)^2\} \right|_0 &= \sum_{i,j} \left. \frac{dp_{ij}(\varepsilon)}{d\varepsilon} \right|_0 \ell_P(i, j)^2 \\ &\quad + 2 \sum_{i,j} p_{ij} \ell_P(i, j) \left. \frac{d\ell_{P_\varepsilon}(i, j)}{d\varepsilon} \right|_0. \end{aligned} \quad (4.24)$$

The first sum is

$$\ell_P(x, y)^2 - \mathbb{E}_P\{\ell_P(X, Y)^2\}. \quad (4.25)$$

For the second sum, define the probability-weighted row and column means

$$\mathbb{E}_P\{\ell_P(X, Y) \mid X = x\} = \frac{\sum_j p_{xj} \ell_P(x, j)}{p_{x+}}, \quad (4.26)$$

$$E_P\{\ell_P(X, Y) \mid Y = y\} = \frac{\sum_i p_{iy} \ell_P(i, y)}{p_{+y}}. \quad (4.27)$$

Substitution of Equation (4.15) gives

$$\begin{aligned} & \sum_{i,j} p_{ij} \ell_P(i, j) \left. \frac{d\ell_{P_\varepsilon}(i, j)}{d\varepsilon} \right|_0 \\ &= \ell_P(x, y) - E_P\{\ell_P(X, Y) \mid X = x\} - E_P\{\ell_P(X, Y) \mid Y = y\} + I(P). \end{aligned} \quad (4.28)$$

Combining Equations (4.18), (4.22), (4.25), and (4.28) yields

$$\boxed{\begin{aligned} g_P(x, y) &= \{\ell_P(x, y) - I(P)\}^2 - V(P) \\ &\quad + 2\left[\ell_P(x, y) - E_P\{\ell_P(X, Y) \mid X = x\} \right. \\ &\quad \left. - E_P\{\ell_P(X, Y) \mid Y = y\} + I(P)\right]. \end{aligned}} \quad (4.29)$$

The first line is the change if the pointwise-MI values were held fixed. The bracket accounts for changes in those values, including the effects of the selected joint probability and both margins.

4.3 Component degrees of freedom and final test

Population component degrees of freedom.

Define the population variability of the observation-level sensitivities by

$$\tau^2(P) = \text{Var}_P\{g_P(X, Y)\}. \quad (4.30)$$

The mean of an influence function is zero. Centring explicitly gives the same variance if numerical substitution produces a small nonzero mean.

Assumption 5 (Regular variance sensitivity). *For the first-order Expanded Welch construction, $\tau^2(P) > 0$ and $\tau^2(Q) > 0$.*

A first-order Taylor expansion of the variance functional around P is

$$\hat{V}(P) - V(P) \approx \frac{1}{n_P} \sum_{a=1}^{n_P} g_P(Z_a^{(P)}). \quad (4.31)$$

The observations are independent, so

$$\text{Var}\{\hat{V}(P)\} \approx \frac{1}{n_P^2} \sum_{a=1}^{n_P} \text{Var}_P\{g_P(Z_a^{(P)})\} \quad (4.32)$$

$$= \frac{\tau^2(P)}{n_P}. \quad (4.33)$$

Substitution into Equation (4.7) gives the population component degree of freedom

$$\boxed{\nu_P \approx \frac{2n_P V(P)^2}{\tau^2(P)}}. \quad (4.34)$$

Equation (4.34) depends on relative, not absolute, uncertainty. Since

$$\frac{\text{Var}\{\hat{V}(P)\}}{V(P)^2} \approx \frac{\tau^2(P)}{n_P V(P)^2}, \quad (4.35)$$

the component degrees of freedom are approximately twice the reciprocal of this relative variance. A stable estimate relative to the size of $V(P)$ has large degrees of freedom; a relatively unstable estimate has smaller degrees of freedom and produces a heavier-tailed final reference. This first-order component formula assumes $\tau^2(P) > 0$. Positive $V(P)$ alone does not guarantee that condition: at a stationary point of the variance functional, its first-order sensitivity can vanish even though MI and $V(P)$ are positive. There, a higher-order analysis of $\hat{V}(P)$ would be needed to describe its finite-sample distribution.

Table-based component degrees of freedom.

The observed table replaces every population quantity in Equation (4.29) by its empirical counterpart. The sums below are over occupied cells, because pointwise MI is undefined at an empirical zero. For each occupied cell,

$$\begin{aligned} \hat{g}_P(i, j) = & \{\hat{\ell}_P(i, j) - \hat{I}(P)\}^2 - \hat{V}(P) \\ & + 2 \left[\hat{\ell}_P(i, j) - \frac{\sum_{j'} \hat{p}_{ij'} \hat{\ell}_P(i, j')}{\hat{p}_{i+}} - \frac{\sum_{i'} \hat{p}_{i'j} \hat{\ell}_P(i', j)}{\hat{p}_{+j}} + \hat{I}(P) \right]. \end{aligned} \quad (4.36)$$

In all weighted sums, an empty cell contributes zero; no logarithm or sensitivity needs to be evaluated there. The empirical variance is centred explicitly:

$$\hat{\tau}^2(P) = \sum_{i,j} \hat{p}_{ij} \left\{ \hat{g}_P(i, j) - \sum_{k,l} \hat{p}_{kl} \hat{g}_P(k, l) \right\}^2. \quad (4.37)$$

The estimated component degree of freedom is

$$\boxed{\hat{\nu}_P = \frac{2n_P \hat{V}(P)^2}{\hat{\tau}^2(P)}}. \quad (4.38)$$

Repeating the same calculation for the second table gives $\hat{\nu}_Q$.

Combining the two components.

Substituting the two variance contributions and component degrees of freedom into Equation (4.1) gives

$$\hat{\nu}_{\text{expanded}} = \frac{\left\{ \hat{V}(P)/n_P + \hat{V}(Q)/n_Q \right\}^2}{\left\{ \hat{V}(P)/n_P \right\}^2 / \hat{\nu}_P + \left\{ \hat{V}(Q)/n_Q \right\}^2 / \hat{\nu}_Q}. \quad (4.39)$$

Substituting Equation (4.38) and its Q analogue gives the algebraically equivalent form

$$\hat{\nu}_{\text{expanded}} = \frac{\left\{ \hat{V}(P)/n_P + \hat{V}(Q)/n_Q \right\}^2}{\hat{\tau}^2(P)/(2n_P^3) + \hat{\tau}^2(Q)/(2n_Q^3)}. \quad (4.40)$$

Equation (4.40) remains defined when one table has $\hat{V} = \hat{\tau}^2 = 0$ and the other supplies a positive denominator contribution. The zero table drops out, so the combined degree of freedom reduces to the other table's component. The implementation evaluated in this thesis does not use the direct form in that case: it first requires both component degrees of freedom in Equation (4.38) to be positive and finite. Reported Expanded-only invalidity can therefore reflect this rule rather than a failure of the combined Satterthwaite formula.

The expanded two-sided p-value is

$$p_{\text{expanded}} = 2 \left\{ 1 - F_{t_{\hat{\nu}_{\text{expanded}}}}(|T|) \right\}, \quad (4.41)$$

where F_{t_ν} denotes the cumulative distribution function of a Student t variable with ν degrees of freedom. As with Wald, only test behaviour is evaluated; no confidence-interval coverage claim is made.

The expanded calculation does not change $\hat{\Delta}_{\text{BC}}$, $\widehat{\text{SE}}$, or T . It changes only the quantile used to evaluate that statistic.

Proposition 1 (Nested rejection under the implemented rules). *If the two calculations use the same finite statistic T and Expanded Welch has positive finite degrees of freedom, its two-sided p-value is at least the Normal Wald p-value. The implemented Expanded-valid set is a subset of the Wald-valid set. Thus, at a common fixed threshold, every Expanded Welch rejection is also a Wald rejection, both replicate by replicate and in the unconditional rates that count invalid outputs as non-rejections.*

Proof. Write a Student variable as $Z/\sqrt{S}$, with independent $Z \sim N(0, 1)$ and $S \sim \chi_\nu^2/\nu$. For $c > 0$, the conditional two-sided tail $2\{1 - \Phi(c\sqrt{s})\}$ is strictly convex

in s , because its second derivative is $c\phi(c\sqrt{s})(s^{-3/2} + c^2s^{-1/2})/2 > 0$. Since $\mathbb{E} S = 1$, Jensen's inequality gives $\Pr(|Z/\sqrt{S}| > c) > \Pr(|Z| > c)$. At $c = 0$ both tails equal one. $\square$

Invalid Expanded outputs count as non-rejections in the reported unconditional rates. Because the correction changes only the reference, it cannot raise rejection where Wald is already too conservative.

4.4 Worked example and implementation

Consider the two observed 3×3 tables

$$N^P = \begin{pmatrix} 35 & 10 & 5 \\ 10 & 20 & 5 \\ 5 & 5 & 10 \end{pmatrix}, \quad N^Q = \begin{pmatrix} 30 & 5 & 5 \\ 5 & 30 & 5 \\ 5 & 5 & 10 \end{pmatrix}. \quad (4.42)$$

Their totals are $n_P = 105$ and $n_Q = 100$. The plug-in MI estimates are 0.13693 and 0.25848 nats. Since $(3 - 1)(3 - 1) = 4$, the corrected difference is

$$\hat{\Delta}_{\text{BC}} = \left(0.13693 - \frac{4}{2(105)}\right) - \left(0.25848 - \frac{4}{2(100)}\right) = -0.12059. \quad (4.43)$$

The empirical influence variances are $\hat{V}(P) = 0.27189$ and $\hat{V}(Q) = 0.43371$. Hence

$$\widehat{\text{SE}} = \left(\frac{0.27189}{105} + \frac{0.43371}{100}\right)^{1/2} = 0.08323, \quad T = -1.44898. \quad (4.44)$$

Normal Wald gives $p = 0.14734$.

For Expanded Welch, the table-based sensitivity calculation first gives $\hat{\tau}^2(P) = 0.93212$ and $\hat{\tau}^2(Q) = 0.82035$. Substitution into Equation (4.38) gives component degrees of freedom

$$\hat{\nu}_P = 16.65, \quad \hat{\nu}_Q = 45.86. \quad (4.45)$$

Equation (4.39) combines them into $\hat{\nu}_{\text{expanded}} = 59.03$, giving $p = 0.15264$. Both methods fail to reject at the 5% level for these tables. The example shows that Expanded Welch changes the reference, not the observed statistic.

Operational algorithm and complexity.

Table 4.1 summarises one comparison.

Each empirical probability, pointwise-MI value, row mean, column mean, and

Table 4.1: Operational Expanded Welch–Satterthwaite algorithm.

Step	Calculation
1	Validate the two aligned count tables and their declared alphabet dimensions; independence of samples is a design assumption.
2	Compute empirical probabilities, pointwise MI, plug-in MI, and the leading bias-corrected difference.
3	Compute $\hat{V}(P)$, $\hat{V}(Q)$, the shared standard error, and T .
4	Compute the row and column pointwise-MI means needed in $\hat{g}_P$ and $\hat{g}_Q$.
5	Compute $\hat{\tau}^2(P)$, $\hat{\tau}^2(Q)$, and the two component degrees of freedom.
6	Combine the components using Equation (4.39).
7	Return the Student p-value, effective degrees of freedom, and validity diagnostics.

variance contribution is calculated by a fixed number of table scans. The time complexity is therefore $O(rc)$, and the working memory is also $O(rc)$. The method is deterministic once the two count tables are given.

4.5 Behaviour and limitations

Near independence.

The size of the adjustment has a useful local explanation. For a distribution close to independence, a second-order Taylor expansion gives

$$V(P) \approx 2I(P), \quad \tau^2(P) \approx 4V(P). \quad (4.46)$$

Substitution into the component formula then gives

$$\nu_P = \frac{2n_P V(P)^2}{\tau^2(P)} \approx n_P I(P). \quad (4.47)$$

Thus low MI can produce a small effective degree of freedom and a much heavier-tailed reference. The approximation is local: it assumes fixed table dimension, positive probabilities and sufficiently regular cell counts. Very near independence, quadratic sampling variation can still be important, so a first-order Student adjustment need not improve finite-sample calibration.

Theoretical status and validity conditions.

The influence-function calculations in Equations (4.18) and (4.29) are directional derivatives of the corresponding population functionals. Under fixed positive support,

they provide the usual first-order asymptotic linear representations.

The scaled chi-squared model in Equation (4.2) and the final Student reference are working approximations. They match the first two moments of the estimated variance and do not provide an exact finite-sample law. Moreover, $\hat{I}(P)$ and $\hat{V}(P)$ use the same observations; their first-order covariance is generally $E_P[\{\ell_P(X, Y) - I(P)\}g_P(X, Y)]/n_P$, not zero. Moment matching the denominator does not establish independence from the numerator or an exact Student law. Under regular large-sample conditions with positive variance sensitivities, $\hat{\nu}_P$, $\hat{\nu}_Q$, and $\hat{\nu}_{\text{expanded}}$ increase, so the Student reference approaches the same normal limit as Wald inference.

The software first requires matching table shapes, at least two rows and two columns, finite nonnegative integer counts, and more than one observation in each table. Wald is then valid only when the corrected difference and standard error are finite and $\hat{V}(P) + \hat{V}(Q) > 10^{-14}$. Expanded Welch additionally requires positive finite component and combined degrees of freedom. This is the strict component-validity convention stated after Equation (4.40); the software does not evaluate that direct combined form when a component degenerates. Failed conditions are recorded as invalid calculations. The validation in Chapters 5 and 6 evaluates how often these conditions occur and whether the finite-sample correction improves calibration.

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